

Jingxuan Wang

wjx1209457769@gmail.com | 682-283-2222 | Arlington, TX | github.com/Sarahbell1209 | linkedin.com/in/jingxuanwang2002/

Education

University of Texas at Arlington — Arlington, TX

Expected Dec 2026

M.S. in Software Engineering

Beijing Forestry University — Beijing, China

Sep 2021 – Jun 2025

B.E. in Network Engineering

Skills

- **Languages:** Python, TypeScript, Java, SQL
- **Technologies:** React, Django REST Framework, Flask, PyTorch, YOLOv8, OpenCV, Whisper, Gemini, Ollama, AWS, Google Cloud, Docker, Vercel, MySQL, SQLite, Pydantic, Git, REST APIs
- **Spoken Languages:** Chinese (Native), English (Proficient)

Experience

Research Software Developer — Clinical AI Assistant

2026 – Present

University of Texas at Arlington — Arlington, TX

- Built a clinical documentation workflow that takes recorded conversations through transcription, SOAP note generation, clinician review, and write-back to Open Dental.
- Developed the full-stack application with React, TypeScript, and Django REST Framework, implementing workflow state management for audio upload, transcription correction, note review, and clinician approval.
- Integrated Google Cloud Speech-to-Text, Deepgram, and OpenAI Whisper into a multi-provider speech pipeline supporting English, Chinese, Spanish, and mixed-language clinical conversations.
- Used Gemini to generate draft clinical notes and added clinician review plus transcript-based Q&A so users could verify information against the original conversation.
- Integrated the Open Dental Cloud Test API to retrieve patient context and write approved clinical notes back into the dental workflow.
- Containerized services with Docker Compose and deployed the frontend on Vercel and backend services on AWS.

Projects

Creator Distiller — Multimodal Content Analysis Pipeline

Jun. 2026 - Present

Python, Whisper, Ollama, Pydantic, SQLite, Browser Automation

- Built a pipeline to collect public creator videos, transcribe audio, analyze visual and dialogue content, and compare patterns across multiple videos.
- Developed a browser-based collection workflow that parsed and deduplicated 125 public video records from a real creator profile during bounded collection runs.
- Integrated local Whisper speech recognition and achieved approximately 0.26× real-time factor (RTF) in local transcription testing.
- Added citation checks so generated findings can only reference known dialogue turns, semantic beats, analysis items, or evidence IDs; invalid findings are rejected before later analysis.
- Added checkpoints between collection, transcription, multimodal analysis, and evidence validation so each stage can be tested and verified independently.

Wildlife Recognition Device — Raspberry Pi Edge Vision System

Mar. 2025 – Jun. 2025

Python, YOLOv8, OpenCV

- Developed an edge-based wildlife recognition system using YOLOv8 and OpenCV for real-time image capture, preprocessing, object detection, and on-device inference.
- Built a continuous video-processing pipeline achieving approximately 10 FPS and 80%+ detection accuracy on resource-constrained Raspberry Pi hardware.
- Implemented local detection filtering and event logging to retain relevant observations while reducing unnecessary storage consumption.
- Optimized the system for offline edge deployment, enabling continuous inference without persistent cloud connectivity.